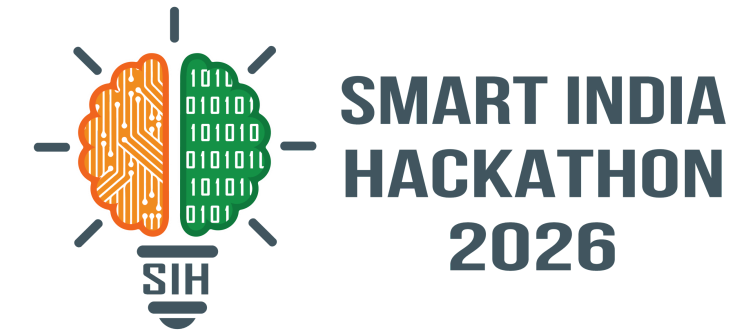
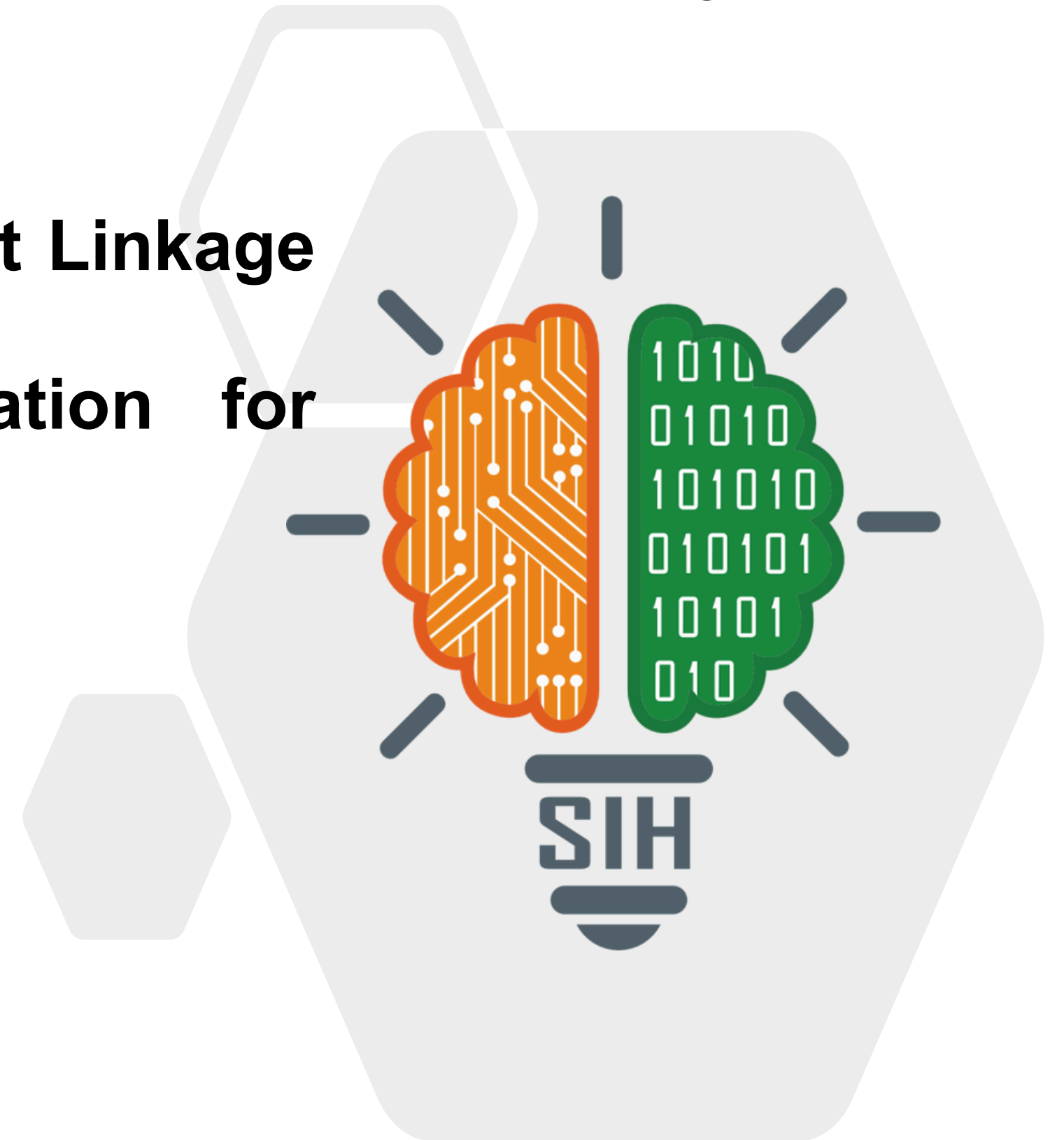


SMART INDIA HACKATHON 2026



- **Problem Statement ID:** SIH26090
- **Problem Statement Title:** AI-Driven Market Linkage and Smart Cataloging Mobile Application for Marginalized Artisans
- **Theme:** Heritage & Culture
- **PS Category:** Software
- **Team ID:**
- **Team Name:** Kalpataru



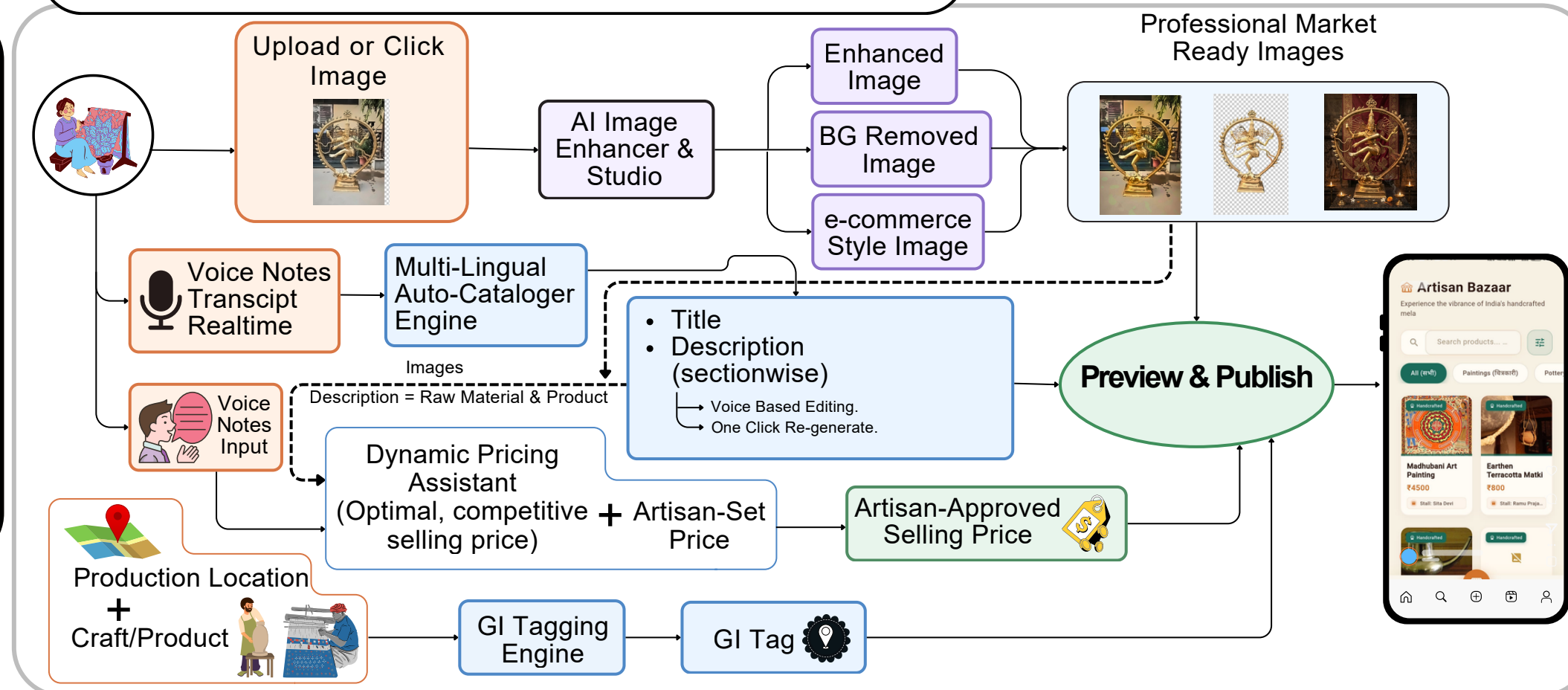
PROPOSED SOLUTION:

- **AI Product Photography:** Enhances images, removes backgrounds, and creates **marketplace-ready photos**.
- **Voice-First Multilingual Cataloging:** Converts regional-language voice input into **professional product listings**.
- **AI Product Understanding:** Identifies product type, material, color, style, and category.
- **Smart Pricing Assistant:** Suggests competitive prices using product, market, and raw-material data.
- **One-Tap Digital Storefront:** Combines product details into a **ready-to-publish catalog**.
- **Direct Market Linkage:** Connects artisans with **B2B, institutional, government buyers** and **individual customers**.

HOW IT ADDRESSES THE PROBLEM :

- **Language & Literacy Barriers:** Voice-first, **28 Multilinguistic interaction** removes the need for complex forms and typing.
- **Undervaluation & Middlemen:** Explainable pricing uses material, labour and packaging costs + market evidence to support fair pricing.
- **Poor Product Presentation:** AI enhances lighting/backgrounds while preserving craft authenticity.
- **Loss of Artisan Identity:** Provenance-aware catalogs preserve artisan facts, craft stories and verified research separately.
- **Limited Market Access:** Connects approved products with buyers, B2B opportunities and government marketplaces.
- **Rural Connectivity:** Offline drafts and synchronization allow work to continue during poor network conditions.

WORKFLOW DIAGRAM:



INNOVATION AND UNIQUENESS OF THE SOLUTION :

- **Interactive Video Registration:** Guided video onboarding helps artisans create their digital identity without complex forms.
- **Voice-First Interaction:** Voice-based navigation and product creation minimize typing and technical complexity.
- **One-Click Multi-Marketplace Publishing:** Publish a single catalog across platforms like Amazon, Flipkart, Meesho, etc.
- **Digital Artisan Community:** Connect artisans to share knowledge, collaborate, discover opportunities and grow together.
- **Integrated Raw-Material Marketplace:** Connect artisans directly with suppliers offering quality materials at reasonable prices.

TECH STACK:

Prototype

FRONTEND



BACKEND



DATABASE



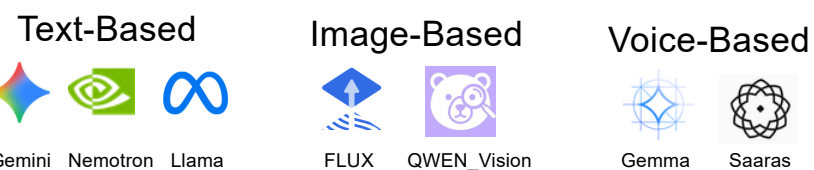
CLOUD



HOSTING



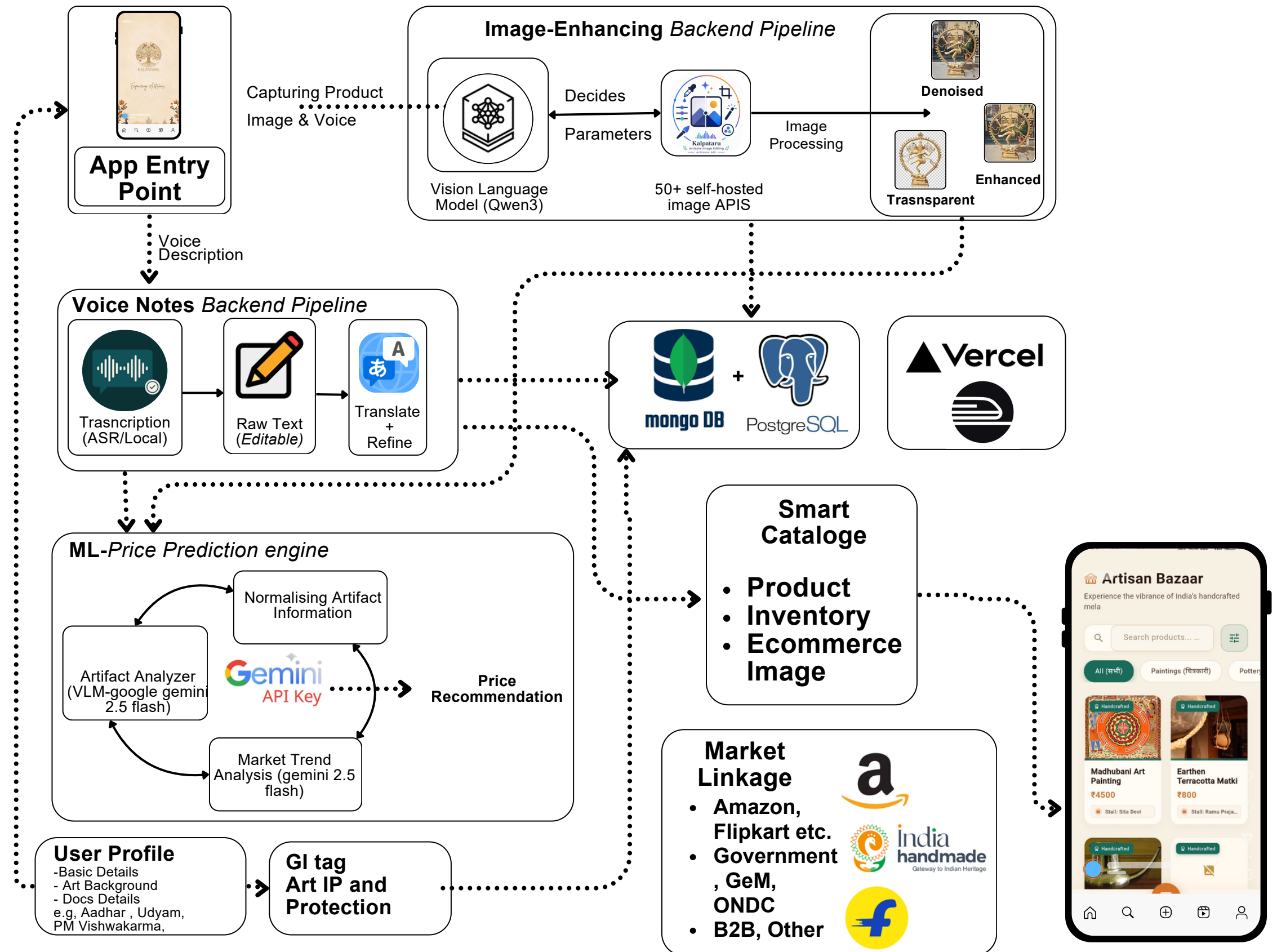
LLM MODELS



LLM- Aggregators/Providers



ARCHITECTURE DIAGRAM :



 TECHNICAL
FEASIBILITY

Flutter-based cross-platform architecture with modular, provider-independent AI integration for ASR, translation, LLM & image services, validated through a working end-to-end prototype.

 OPERATIONAL
FEASIBILITY

Offline-tolerant, asynchronously processed, modular architecture with decoupled media services and scalable cluster-wise deployment.

 ECONOMIC
FEASIBILITY

Low infrastructure lock-in through cross-platform development, open-source components, usage-scalable AI, and provider abstraction for cost-performance optimization.

 SCALABILITY

Modular, non-monolithic architecture enabling independent expansion of artisans, languages, products, AI providers, storage, and market-channel connectors.

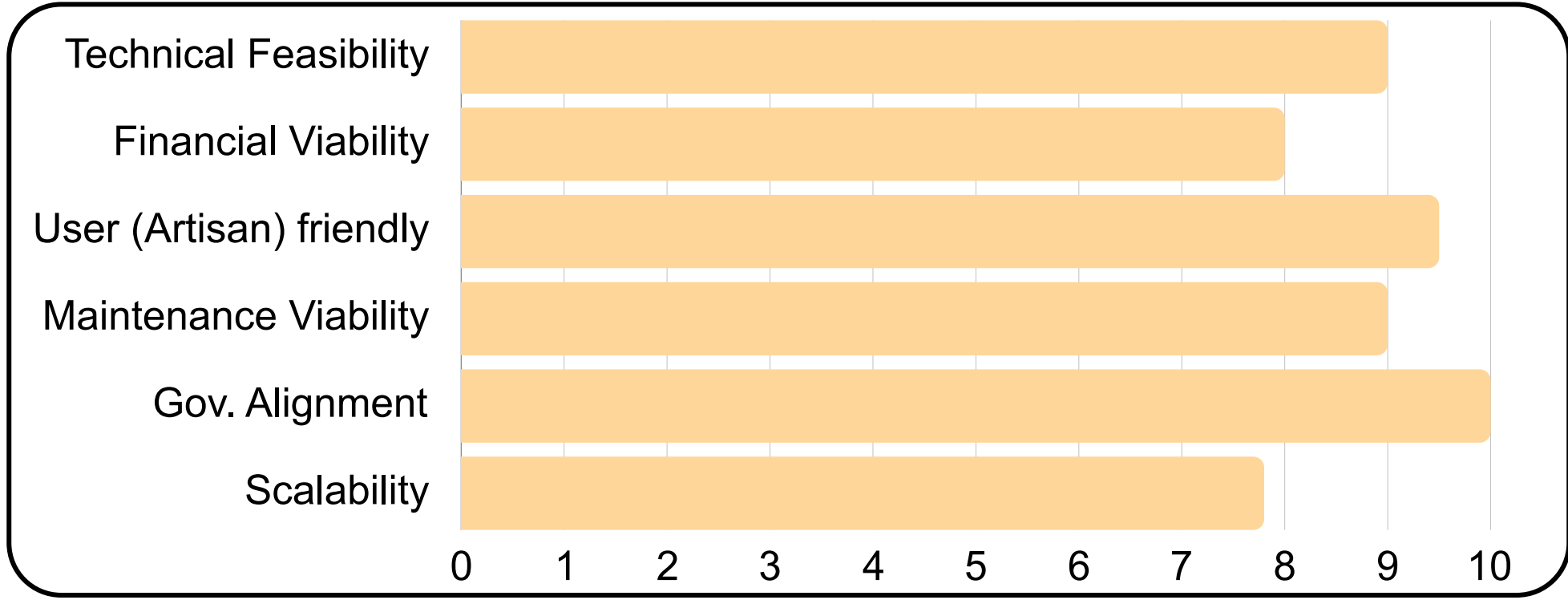
 USER
VIABILITY

Low-literacy-first, voice-driven regional-language UX with minimal typing, visual guidance, task-focused flows, and human-verified publishing

 FINANCIAL
VIABILITY

Multi-sided ecosystem monetization through premium AI & analytics, B2B buyer discovery, raw-material commerce, and institutional/cluster deployments.

Challenges	Strategy
Dialect & Noise Variations in Rural Speech	Real-time audio playback + visual transcript confirmation; ASR confidence threshold > 0.80 triggers manual/audio correction.
AI Hallucinations in Heritage Storytelling	Strict structural separation of artisanProvidedFacts and researchFacts; unverified claims are rejected.
Image Alteration / Loss of Handmade Texture	Automated artifact-fidelity check (>0.93 threshold); original raw photo is immutably archived.
Artisan Skepticism & Distrust of Pricing	Explainable price breakdown showing raw materials + labor wages + market position; artisan retains 100% final pricing control.



POTENTIAL IMPACT ON THE TARGET AUDIENCE

ECONOMIC IMPACT

- **Fairer Income** — Transparent, cost-based pricing.
- **Direct Markets** — Buyers + B2B access.
- **Lower Selling Cost** — Simple AI-powered listing.
- **Year-Round Sales**: Beyond fairs & exhibitions.

SOCIAL IMPACT

- **Digital Inclusion** — Voice-first & multilingual
- **Artisan Independence** — Create, price & manage
- **Greater Dignity** — Artisan stays in control
- **Bridging the Digital Divide** — Accessible commerce

CULTURAL IMPACT

- **Heritage Preservation** — Stories, skills & lineage.
- **Authentic Representation** — Craft fidelity protection.
- **Digital Provenance** — Traceable artisan identity.
- **Cultural Connection** — Story-driven buying.

POLICY IMPACT

- **Scheme Access** — PM Vishwakarma, ODOP, MUDRA
- **Market Linkage** — Government & institutional channels.
- **Craft Intelligence** — Anonymized demand & pricing data.
- **Data-Driven Policy** — Better cluster planning.



ARTISANS

BENEFITS OF THE SOLUTION

IMPACT AREA	KEY BENEFITS	RESULT
Economic 	Fair Pricing, Direct Sales, Wider Markets and B2B Opportunities	Higher & Sustainable Income
Social 	Digital Inclusion, Voice-First Access and Artisan Independence	Dignified Livelihoods
Cultural 	Heritage Preservation, Provenance, Craft Storytelling and Authenticity	Living Cultural Legacy
Policy 	Scheme Access, Government Linkage, Market Insights and Cluster Data	Data-Driven Craft Development

FUTURE SCOPE

1. AR Craft Discovery: Let customers use Augmented Reality to visualize handcrafted products in their spaces while simultaneously exploring the artisan and cultural story behind them.
2. Enable deeper B2B and government-marketplace integrations, while introducing advanced analytics and AI-assisted demand and pricing recommendations.
3. Expand to global markets with international buyer connections, AI-based demand forecasting, and richer digital storytelling for Indian crafts.
4. Automated integration with GeM and ONDC, onboarding 50,000+ artisans via cluster kiosks.

SELF-SUSTAINING BUSINESS MODEL

Artisan Core (Free): Zero-cost voice cataloging, image studio, and local marketplace listing.

Buyer Transaction Fee: 2–4% platform convenience fee strictly on successful buyer checkouts.

B2B Bulk Matchmaking: Subscription & commission tier for verified bulk sourcing by hotels, retailers, and exporters.

Value-Added Studio Services: Premium packaging, logistics linkage and promoted catalog positioning.

Government / Institutional Analytics: Anonymized cluster demand and pricing intelligence reports for policy planning.

RESEARCH VALIDATION

- Frontiers in Human Dynamics (2026): Digital platforms can expand artisan product visibility, discoverability and access to global markets. [Link](#)
- E-Commerce & Indigenous Artisans (2025): Digital illiteracy, logistics and mass-produced competition limit artisans from benefiting fully from e-commerce. [Link](#)
- Catalyst Impact Accelerator: 98% of surveyed artisans had never made an online sale; 57% needed help uploading photos and 60% navigating platforms highlighting the usability gap. [Link](#)
- 545 Handicraft Enterprises Study (UP): 26.6% lacked e-commerce knowledge, while legal and economic barriers kept many businesses confined to local markets. [Link](#)
- Pan-India Marketplace Study: Fragmented markets and intermediary dependence continue to limit direct artisan-to-consumer access. [Link](#)

Prototype: [LINK](#)

Field Survey & Validation: [Link](#)

Product Demo Video: [Link](#)

COMPETITIVE FEATURE GAP ANALYSIS

What the SIH PS demands	Kalpataru	EcomCatalog AI	India Handmade	Gaatha
Voice-First Regional Cataloging	✓	✓	×	×
AI Craft Image Studio (Fidelity > 0.93)	✓	×	×	×
Craft-aware dynamic price recommendation	✓	×	×	×
Automated GI Tagging & Geo-Provenance	✓	×	✓	×
Offline-Tolerant Mobile Workflow	✓	×	△	✓
Direct MoSJE / GeM / ODOP Scheme Bridge	✓	△	△	×
Product + artisan story + cultural identity	✓	×	△	✓